A Comparative Analysis of StandardScaler and MinMaxScaler Normalization for Deep Neural Network-Based Dengue Fever Classification

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***Abstract*—Dengue Hemorrhagic Fever (DHF) remains a major public health challenge in tropical and subtropical regions, requiring rapid and accurate diagnosis to minimize the risk of complications and mortality. One of the challenges in deep learning-based classification is the variability in clinical feature values, which can affect the model's learning process. This study aims to compare the effects of three normalization methods, no normalization, StandardScaler, and MinMaxScaler—on the performance of a Deep Neural Network (DNN) model using two dataset configurations: a complete dataset (clinical symptoms and laboratory results) and a dataset of clinical symptoms for initial screening scenarios. To ensure the validity of the evaluation, the processes of imputing missing values and normalization were performed only on the training data during each stage of 5-fold cross-validation, thereby avoiding data leakage. The final model was then tested using a separate test dataset. The results of this study show that normalization significantly improves classification performance. StandardScaler yielded the best results on the complete dataset, with an accuracy of 94.30%, an F1 score of 94.42%, and an ROC-AUC of 0.9923. These findings indicate that selecting the appropriate normalization method is crucial for improving the reliability of DNN models in classifying dengue fever using diverse clinical data.**

***Index Terms*—Dengue Hemorrhagic Fever, Deep Neural Network, Data Normalization, StandardScaler, Clinical Data Classification**

I. Introduction

Dengue Hemorrhagic Fever (DHF) remains a significant health challenge in tropical and subtropical regions, with cases increasing annually. The World Health Organization (WHO) estimates that there are approximately 100–400 million cases of dengue fever worldwide each year, making rapid and accurate diagnosis crucial to reduce the risk of complications and death [1]–[3]. Dengue fever is typically diagnosed by combining clinical symptoms with laboratory test results, including platelet count, leukocyte count, hematocrit, and hemoglobin count [4]. However, dengue symptoms often mimic those of other febrile illnesses, complicating the diagnosis process, especially in the early stages of infection. To address these challenges, artificial intelligence (AI), particularly in the fields of Machine Learning (ML) and Deep Learning (DL), is increasingly being used for disease classification. Among the various available methods, Deep Neural Networks (DNNs) have demonstrated strong performance in analyzing structured clinical data [5]–[9].

However, the performance of deep learning models is significantly influenced by data preprocessing, particularly feature normalization. Clinical datasets typically consist of numeric attributes with varying value ranges, which can impact the model's learning process if not managed properly [10]. Furthermore, normalization performed before data segmentation can lead to data leakage and overly positive assessments. Therefore, this study analyzes three normalization techniques: no normalization, StandardScaler, and MinMaxScaler, in the context of a DNN model with two dataset types: a full dataset (clinical symptoms and laboratory results) and a dataset containing only clinical symptoms for initial screening scenarios. All imputation and normalization procedures were performed only on the training data at each stage of Stratified 5-Fold Cross Validation to prevent data leakage, while the final evaluation was conducted using separate test data. Model performance was then analyzed using metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC [11].

II. Related Work

Previous research has shown that both Machine Learning (ML) and Deep Learning (DL) are effective approaches for dengue classification. A systematic review by Andrade Girón et al. [12] highlighted the successful application of algorithms such as Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Random Forest, Support Vector Machine (SVM), and Extreme Gradient Boosting (XGBoost). The review also emphasized that data preprocessing, particularly feature normalization and K-Fold Cross Validation, plays a key role in improving classification performance and model generalization. Similarly, previous research has shown that normalization methods, including Z-score and Min-Max Scaling, along with cross validation, contribute to higher classification accuracy and more reliable model performance [13]–[17].

Several dengue classification studies have reported accuracy above 97% using ANN and SVM models with appropriate preprocessing and validation strategies [18]–[21]. Other comparative studies achieved accuracy exceeding 99% using feature normalization combined with machine learning algorithms such as Extra Tree Classifier and Decision Tree [3], [22]. Similar findings have also been reported in Indonesia, where Min-Max Scaling improved the performance of an XGBoost-based dengue classification model [5]. Overall, these studies show that feature scaling and validation strategies substantially impact model accuracy and generalization [23].

Although previous research has demonstrated the benefits of normalization, most studies have only evaluated one preprocessing technique or one dataset configuration. Comprehensive comparisons of different normalization methods for DNN-based dengue classification across different dataset configurations are rare. To address this limitation, this study systematically compares three normalization strategies using complete clinical-laboratory data and symptom-only datasets.

III. Methodology

A. Research Workflow

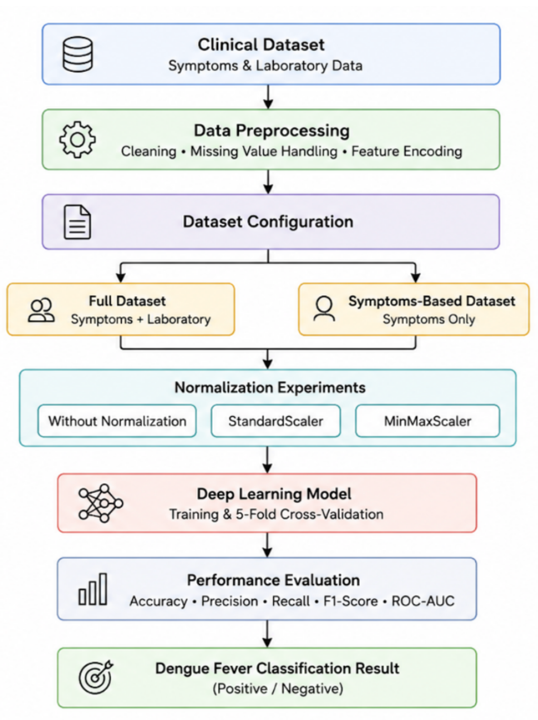

Fig. 1. Research Workflow

Fig. 1 illustrates the overall workflow of this study. Clinical symptom and laboratory data were collected and preprocessed through data cleaning, missing value imputation, and feature encoding. The processed data were then divided into two dataset configurations: a complete dataset and a symptoms-only dataset. Each configuration was evaluated using three normalization approaches (no normalization, *StandardScaler*, and *MinMaxScaler*) before being used to train a Deep Neural Network (DNN) with stratified 5-Fold Cross Validation. Finally, model performance was assessed using Accuracy, Precision, Recall, F1-Score, and ROC-AUC to identify the most effective normalization method for dengue fever classification.

B. Dataset

The dataset used in this study consists of clinical records collected from patients presenting with febrile illness. Each record includes demographic information, vital signs, clinical symptoms, laboratory examination results, and a diagnosis label indicating either Dengue Fever or non-dengue fever. In total, the dataset contains 21 input features and one target label, as summarized in Tables I and II.

TABLE I
Input Feature Categories

Category	Attributes	Experiments Applied
Demographic Data	Age, Gender	Age Only
Vital Signs	Temperature, Systolic Blood Pressure, Diastolic Blood Pressure	Yes
Laboratory Features	Platelet, Leukocyte, Hematocrit, Hemoglobin	Yes
Clinical Symptoms	Fever, Fever Duration, Headache, Nausea, Vomiting, Abdominal Pain, Fatigue, Loss of Appetite, Red Spots, Body Pain, Abdominal Distension, Joint Pain	Fever Duration Only

TABLE II
Target Label Description

Target Label	Description
0	Fever
1	Positive Dengue Fever

This dataset consists of 1,141 patient medical records collected from Permata Gading Clinic between January 2024 and December 2025. The class distribution is relatively balanced, consisting of 575 non-dengue cases and 566 dengue cases; therefore, no additional resampling techniques, such as SMOTE, were applied. Normalization was performed only on continuous numeric features, including age, body temperature, systolic and diastolic blood pressure, laboratory measurements, and fever duration. Binary variables representing clinical symptoms, gender, and target label were excluded from the normalization process because their values already represent categorical information.

C. Data Preprocessing

The data preparation phase involved eliminating duplicate entries, filling missing values using the median for numerical features and the most frequently occurring values for categorical features, and converting categorical variables to binary values. All 21 features were retained as they were directly related to dengue fever diagnosis. To avoid data leakage, the filling and normalization processes were applied only to the training data in each cross-validation section. Next, the dataset was stratified with an 80:20 ratio to form the training-validation and test data sets. The model was built using Stratified 5-Fold Cross Validation, then retrained on the entire training-validation data set with an internal ratio of 85:15 for early stopping, before being tested on the separate test data set to obtain objective results.

D. Normalization Experiments

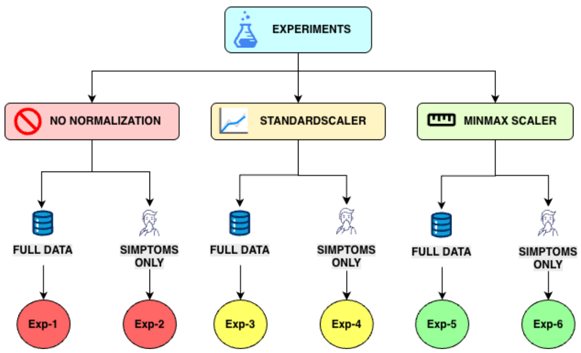

Fig. 2. Normalization Experiment Framework

Fig. 2 illustrates the experimental framework adopted in this study. Two dataset configurations were evaluated: the complete dataset, which includes both clinical symptoms and laboratory examination results, and the symptoms-only dataset, which excludes laboratory features to simulate an early screening scenario. Each dataset configuration was tested using three normalization strategies, namely no normalization, *StandardScaler*, and *MinMaxScaler*, resulting in six experimental scenarios.

StandardScaler standardizes numerical features by transforming them to have zero mean and unit variance,

$$z = \frac{x - \mu}{\sigma}, \tag{1}$$

where μ and σ represent the mean and standard deviation of each feature, respectively [24]. Meanwhile, *MinMaxScaler* rescales feature values into the range of $[0, 1]$ using

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}, \tag{2}$$

where x_{min} and x_{max} denote the minimum and maximum values of each feature [25]. The performance of each experimental scenario was subsequently compared to determine the most effective normalization strategy for DNN-based dengue fever classification.

TABLE III
NORMALIZATION EXPERIMENT CONFIGURATIONS

Experiment	Dataset	Normalization
Exp-1	Complete Dataset	None
Exp-2	Symptoms Dataset	None
Exp-3	Complete Dataset	StandardScaler
Exp-4	Symptoms Dataset	StandardScaler
Exp-5	Complete Dataset	MinMaxScaler
Exp-6	Symptoms Dataset	MinMaxScaler

E. Proposed Deep Neural Network Model

A Deep Neural Network (DNN) was developed to classify dengue fever cases using clinical and laboratory data. The network architecture consists of an input layer, two fully connected hidden layers, two dropout layers, and a single-neuron output layer, as presented in Table IV. The number of input neurons depends on the selected dataset configuration. ReLU activation was applied in the hidden layers to introduce non-linearity, while dropout layers were incorporated to reduce the risk of overfitting during training. A Sigmoid activation function was used in the output layer to produce binary classification probabilities.

TABLE IV
PROPOSED DNN ARCHITECTURE

Layer	Configuration	Activation
Input Layer	Depends on dataset configuration	–
Dense Layer 1	32 Neurons	ReLU
Dropout Layer 1	Rate = 0.2	–
Dense Layer 2	16 Neurons	ReLU
Dropout Layer 2	Rate = 0.1	–
Output Layer	1 Neuron	Sigmoid

The model was trained using the Adam optimizer with the Binary Crossentropy loss function. Training was performed for a maximum of 50 epochs with a batch size of 32 and an initial learning rate of 0.001. Early stopping with a patience value of five epochs was applied to prevent unnecessary training once the validation performance no longer improved. The complete training configuration is summarized in Table V.

TABLE V
TRAINING HYPERPARAMETERS

Parameter	Value
Optimizer	Adam
Loss Function	Binary Crossentropy
Epochs	50
Batch Size	32
Learning Rate	0.001
Early Stopping	Patience = 5
Cross Validation	5-Fold
Decision Threshold	0.4

A decision threshold of 0.4 was adopted instead of the default value of 0.5 to increase the model's sensitivity in identifying positive dengue cases. From a clinical perspective, failing to detect an actual dengue patient may have more serious consequences than generating a false positive result, which can be confirmed through subsequent laboratory examinations. All experiments were implemented using TensorFlow 2.x on a workstation equipped with [hardware specification].

F. Evaluation Metrics

The performance of the proposed DNN model was evaluated using several widely adopted classification metrics, including Accuracy, Precision, Recall, F1-Score, and the Area Under the Receiver Operating Characteristic Curve (ROC-AUC) [26], [27]. Accuracy measures the overall proportion of correctly classified instances and is calculated as

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{3}$$

Precision and Recall were used to assess the model's ability to correctly identify positive dengue cases while minimizing false predictions, and are defined as

$$Precision = \frac{TP}{TP + FP}, \qquad Recall = \frac{TP}{TP + FN} \tag{4}$$

The F1-Score represents the harmonic mean of Precision and Recall, providing a balanced measure of classification performance, particularly when both false positives and false negatives are important.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{5}$$

In these equations, *True Positive* (TP) and *True Negative* (TN) denote correctly classified dengue fever and non-dengue fever cases, respectively, whereas *False Positive* (FP) and *False Negative* (FN) represent incorrect classifications. In addition to these metrics, ROC-AUC was used to evaluate the model's discrimination capability across different decision thresholds. Confusion Matrix analysis was also performed to examine the distribution of correct and incorrect predictions and to compare the classification performance obtained from each normalization strategy and dataset configuration.

IV. Results

A. Model Training Performance

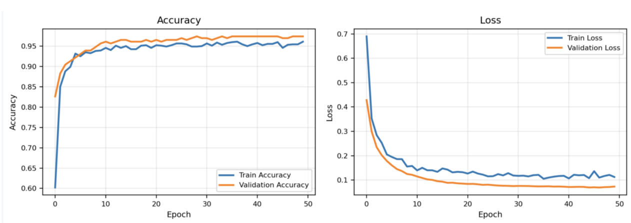

Fig. 3. Training Accuracy and Loss

Fig. 3 presents the learning curves of the proposed DNN model during the training process. Both the training and validation accuracy increased rapidly during the initial epochs before gradually converging to stable values. The validation accuracy reached approximately 97%, while the training accuracy stabilized at around 95–96%, indicating that the model successfully learned the underlying patterns in the dataset.

A similar trend can be observed in the loss curves, where both training and validation loss decreased consistently as training progressed. The validation loss remained slightly lower than the training loss throughout most epochs, which is likely attributable to the use of dropout regularization during training. Furthermore, the small gap between the training and validation curves suggests that the model generalized well to unseen data and exhibited no significant signs of overfitting.

B. Normalization Experiment Results

Six experimental scenarios were conducted by combining two dataset configurations (complete and symptoms-only) with three normalization approaches: without normalization, *StandardScaler*, and *MinMaxScaler*. Table VI presents an example of how numerical feature values changed after each normalization method.

TABLE VI
Comparison of Data Before and After Normalization

Attribute	Unit	Without Normalization	Standard Scaler	MinMax Scaler
Age	Years	26	0.03	0.33
Temperature	°C	36.7	-1.15	0.18
Systolic BP	mmHg	100	0.05	0.40
Diastolic BP	mmHg	70	0.30	0.25
Platelet	cells/μL	139000	-0.33	0.05
Leukocyte	cells/μL	4500	-0.64	0.07
Hematocrit	%	26.7	-1.63	0.29
Hemoglobin	g/dL	13.5	-0.30	0.63
Fever Duration	Days	5	1.43	0.71

TABLE VII
Normalization Experiment Results

Experiment	Accuracy	Precision	Recall	F1-Score
Exp-1	56.58%	100.00%	12.39%	22.05%
Exp-2	76.75%	69.23%	95.58%	80.30%
Exp-3	**95.61%**	**93.28%**	**98.23%**	**95.69%**
Exp-4	88.16%	89.09%	86.73%	87.89%
Exp-5	94.30%	91.67%	97.35%	94.42%
Exp-6	90.35%	88.89%	92.04%	90.43%

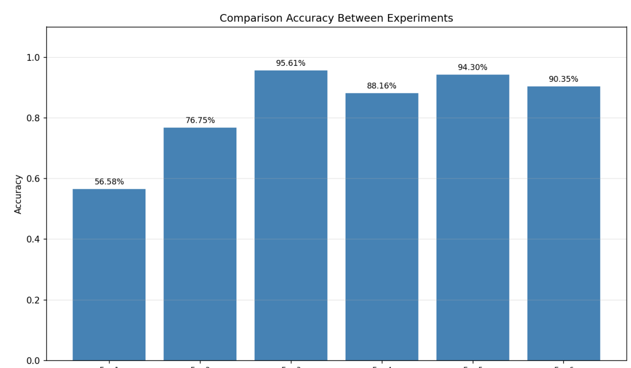

Fig. 4. Comparison Accuracy Between Experiments

Table VII and Figure 4 indicate that feature normalization plays a crucial role in improving the classification effectiveness of the proposed DNN model. Experiments without normalization (Exp-1 and Exp-2) yielded the lowest performance, indicating that the scale variation between features makes it difficult for the model to perceive relevant patterns from the clinical data. In contrast, applying normalization drastically improved the model's ability to identify dengue cases.

Among all evaluated configurations, *StandardScaler* achieved the best overall performance on the complete dataset

(Exp-3), with an accuracy of 95.61%, a precision of 93.28%, a recall of 98.23%, and an F1-score of 95.69%. *MinMaxScaler* also produced competitive results, reaching an accuracy of 94.30% and an F1-score of 94.42% on the same dataset (Exp-5). For the symptoms-only dataset, both normalization methods considerably improved classification performance compared with the experiment without normalization, although the complete dataset consistently achieved higher results.

Overall, the findings of this study underscore that feature normalization is a crucial initial step in dengue fever classification using DNNs and diverse clinical data. Furthermore, using the full dataset consistently yielded superior performance compared to datasets that only included clinical symptoms. This suggests that laboratory test data provides additional information that helps the model more accurately distinguish between dengue and non-dengue fever patients.

C. Classification Evaluation

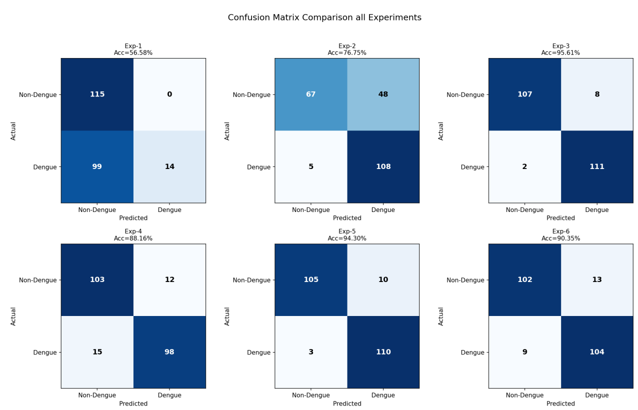

Fig. 5. Confusion Matrix Comparison Across All Experiments

Figures 5 presents the confusion matrices for the six experimental scenarios. The models trained without feature normalization (Exp-1 and Exp-2) produced the highest number of misclassified samples. In particular, Exp-1, which used the complete dataset without normalization, correctly identified only a small proportion of dengue cases, resulting in low recall and overall accuracy. Likewise, Exp-2, which used only symptom data without normalization, generated many false positive predictions, indicating that the model had difficulty distinguishing dengue from non-dengue cases when the input features had different numerical scales.

After feature normalization was applied, classification performance improved considerably. On the complete dataset, *StandardScaler* (Exp-3) produced the most accurate predictions with the fewest misclassified samples, while *MinMaxScaler* (Exp-5) achieved comparable results with only a slight increase in classification errors. For the symptoms-only dataset, both normalization methods (Exp-4 and Exp-6) also reduced the number of misclassifications compared with the corresponding experiment without normalization, demonstrating that normalization helps the DNN learn more representative patterns from the clinical data.

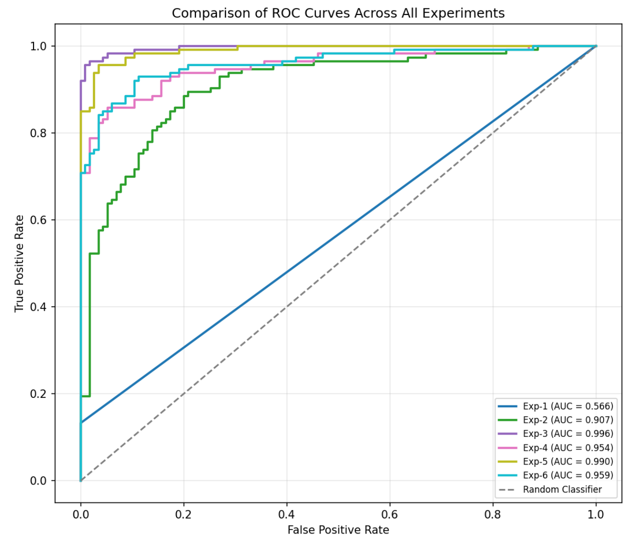

Fig. 6. ROC Curve Comparison Across All Experiments

Figure 6 compares the ROC curves obtained from all six experimental scenarios. The highest discrimination capability was achieved by *StandardScaler* on the complete dataset (Exp-3), with a ROC-AUC of 0.9961, followed by *MinMaxScaler* (Exp-5) with a ROC-AUC of 0.9903. On the symptoms-only dataset, both normalization methods also produced strong discrimination performance, with ROC-AUC values above 0.95. In contrast, the models trained without normalization achieved substantially lower ROC-AUC values, particularly Exp-1, which obtained only 0.5664. These results confirm that feature normalization is an essential preprocessing step for improving the discrimination capability and generalization performance of the proposed DNN model, with *StandardScaler* providing the most consistent overall performance.

V. Discussion

The study results show that feature normalization significantly improves the effectiveness of Deep Neural Network (DNN) models in Dengue Hemorrhagic Fever (DHF) classification. Models without normalization exhibit lower performance, while applying normalization improves all evaluation metrics. Among the various methods tested, *StandardScaler* yields the most optimal results, indicating that standardization of numerical features is more efficient than using raw data or *MinMaxScaler*.

A comparison of dataset configurations shows that the comprehensive dataset, which combines clinical symptoms with laboratory results, consistently outperforms the symptom-only dataset. This indicates that laboratory information plays a crucial role in improving classification accuracy. However, the symptom-based dataset still performed well after normalization, thus retaining its potential as an initial screening tool.

A review of the confusion matrix, ROC curve, and cross-validation showed that the model demonstrated good generalization ability without significant signs of overfitting. The best results were achieved by combining the complete dataset with *StandardScaler*, achieving an accuracy of 95.61% and an ROC-AUC of 0.9961, indicating that the proposed approach is effective and reliable for dengue fever classification based on clinical data.

This study has several limitations. The proposed DNN models were only compared based on various normalization techniques and dataset settings, without using other machine learning algorithms as benchmarks. Furthermore, the dataset used was from a single institution with limited data, so the model's generalizability to more diverse populations needs further evaluation. Therefore, future research needs to validate the model using larger datasets from multiple institutions to ensure its reliability in diverse clinical situations.

VI. Conclusion

This study explores the impact of feature normalization on the effectiveness of Deep Neural Network (DNN) in classifying Dengue Hemorrhagic Fever (DHF) using two settings on a clinical dataset. The findings of this study indicate that the application of feature normalization significantly improves model performance compared to the unnormalized state, where *StandardScaler* provides the highest performance on the complete dataset with an accuracy score of 95.61%, an F1-score of 95.69%, and an ROC-AUC of 0.9961. In addition, the utilization of a dataset that includes clinical symptoms and laboratory test results consistently shows better results compared to a dataset that only contains symptoms. Overall, the proposed framework, which applies data leakage-free preprocessing and Stratified 5-Fold Cross Validation, is proven effective in producing a reliable DHF classification model that has the potential to assist in early detection and clinical decision-making.

VII. Future Work

Future research could test the model using larger and more diverse datasets from multiple healthcare institutions. Furthermore, comparisons with other machine learning algorithms and deep learning architectures, as well as integrating the model into a web-based clinical decision support system, are expected to improve performance and expand the model's use in clinical practice.

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